

On the Quadrature of the Circle

It is a highly remarkable phenomenon that, despite the great advances which mathematics has made in recent centuries - advances that may be said to exceed all expectations - several questions that must have presented themselves already in the earliest times, and that are in fact almost as old as science itself, have still not been decided in spite of all efforts. Among these problems is the quadrature of the circle, which had already attained a high degree of fame among the Greeks because of its difficulties. The ancients sought to solve all problems by pure geometrical construction; and for the solution of the problem in question here, understood in the same sense as the Greek mathematicians understood it, it would therefore be necessary to show by what geometrical construction, from the radius of a given circle, one can derive the side of a square having the same area as that circle. The question then arises whether such a derivation is possible at all. A complete answer requires either the specification of the construction by which this derivation can be performed, or the proof that the side of the square cannot be determined geometrically from the radius. Attempts have often been made to apply to this question the fruitful methods invented by the moderns, which have led to so many results that the ancients, with their limited aids, would have had to regard as unattainable; but so far this has not fully succeeded.

Before I pass to the presentation of the single satisfactory result that has arisen from these efforts, it is necessary to say in a few words under what form the question presents itself when translated into the language usual among modern mathematicians. By the quadrature of the circle one understands the derivation, from the radius of the circle, of the side of the square having the same area as a given circle; and this derivation must be performed by geometrical construction. By this name, as is well known, one denotes arbitrary but finite combinations of the following simple constructions or operations: drawing a straight line through two given points, and describing a circle whose centre and radius are given.

In analytic geometry it is shown in detail that, if one line can be derived from another in the manner just specified, then the ratio of the two lines can be represented by an expression in which, besides the four fundamental operations of arithmetic, only one or more extractions of square roots occur; and conversely, if such an expression is given, one can always find by geometrical construction a line that stands to a given line in the ratio determined by the expression. Thus, if the question is to be decided by calculation, it becomes: can the ratio of the diameter to the side of the square equal to the circle be represented by a finite combination of the operations just named? Instead of that ratio one may also consider the ratio of the diameter to the circumference, since each is very easily derived from the other; in other words, the quadrature and rectification of the circle are essentially one and the same problem. In fact the question is usually posed in the latter form, and to decide it one must investigate whether the ratio of the diameter to the circumference, denoted by π , can be represented by an expression of the kind defined above. Everything that has so far been attempted to answer this question, and especially some investigations of LAMBERT to be discussed presently, has made the impossibility of such a representation, and therefore of the quadrature of the circle, very probable. Yet it would still be very desirable to find a rigorous proof of the impossibility.

For apart from the fact that truths without practical applications - and the theorem just described as highly probable, the impossibility of the quadrature of the circle, must undoubtedly be counted among them - interest us only insofar as we have recognized them as truths by strict reasoning, it has also not rarely happened that what appeared, after repeated fruitless efforts, most probably impossible was finally recognized as possible and actually carried out. A notable example is provided by a problem with which the Greek

mathematicians were already occupied, but which has been solved in full generality only in the most recent time: the division of the circle. Already in EUCLID's Elements one finds the geometrical construction for dividing the circle into 3, 5, and 15 parts, from which still others can be derived by repeated bisection of arcs. All later efforts to accomplish the division in other cases by geometrical construction remained fruitless, and it seemed at last to be generally believed that the cases just mentioned were in fact the only ones in which such a division was possible, until GAUSS showed the contrary in a manner as surprising as it was brilliant.